Bayesian Data Analysis of the German Tank Problem

Theoretical Properties, Simulation Benchmarks, and Real-World Stress-Testing

Inferring Discrete Ceilings, Decision Theory, and Operational Assumption Violations

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Historical Context: Statistical Intelligence in WWII

Objective: Infer the unknown total number of tanks produced, N, using a small sample of observed serial numbers.

1. Espionage Inflation

Conventional military intelligence relied on spy networks and captured prisoners, drastically overestimating production by **4.5–8.2**$\times$ (1,000–1,550 tanks/mo).

2. Order Statistics

Allied statisticians analyzed captured chassis and gearbox serials ($k \approx 5$–50), mathematically estimating production at **169–327** tanks/mo.

3. Speer Ministry Ledgers

Post-war Albert Speer production records confirmed statistical estimates converged within **4%–10%** of actual production.

Production Month	Conventional Spy Est.	Statistical Est. ($\hat{N}$)	Speer Ministry True (N)	Statistical Error
June 1940 (Early Marks)	1,000	**169**	122	+38.5%
June 1941 (Panzer III/IV)	1,550	**244**	271	−**9.9**%
August 1942 (Panzer III/IV)	1,550	**327**	342	−**4.4**%

Core Inferential Objective: *Infer the unknown population ceiling N, moving beyond point estimates to complete uncertainty quantification.*

Mathematical Formulation: Data, Parameter, and Likelihood

Explicitly separating what is observed from the unknown parameter we seek to infer.

The Data-Generating Mechanism

Serials drawn by simple random sampling without replacement:

$$S_1, \ldots, S_k \overset{\text{SRSWOR}}{\sim} \{1, 2, \ldots, N\}$$

- **Observed Data:** $D = \{s_1, \ldots, s_k\}$ with sample size k.
- **Order Statistics:** $1 \leq s_{(1)} < s_{(2)} < \cdots < s_{(k)} = m$.
- **Sample Maximum:** $m = \max(s_1, \ldots, s_k)$.
- **Physical Boundary:** $N \geq m$ is an absolute physical constraint.

Inferential Target & Unordered Likelihood

- **Inferential Target:** The target is the unknown population ceiling $N \in \mathbb{N}_{\geq m}$, *not the individual serial numbers*.
- The serial numbers D are observed data; N is the unknown quantity we want to learn about.

$$P(D \mid N, k) = \frac{1}{\binom{N}{k}} \mathbb{I}(N \geq m)$$

- Likelihood is strictly zero for all $N < m$.
- Sample maximum m captures all sample dependence on N.

Key Insight: *The serial numbers are observed evidence; the population ceiling N is the unknown inferential target.*

Classical Solution: The UMVUE & Its Limitations

Frequentist point estimation achieves repeated-sampling unbiasedness but provides no posterior distribution.

The UMVUE Estimator

By the likelihood ratio criterion for minimal sufficiency (Bahadur, 1954), $M = \max_i S_i$ is complete and minimal sufficient.

$$\widehat{N}_{\mathrm{UMVUE}} = m\left(1 + \frac{1}{k}\right) - 1$$

$$\mathrm{Var}(\widehat{N}_{\mathrm{UMVUE}}) = \frac{(N-k)(N+1)}{k(k+2)}$$

- Unbiased across hypothetical repeated samples ($\mathbb{E}[\widehat{N}] = N$).
- Simple closed-form computation requiring only m and k.

Fundamental Limitations of Classical Approach

- **No Posterior Uncertainty Distribution:** The classical approach constructs a point estimator that is unbiased over repeated samples. It does not provide a posterior distribution describing uncertainty about N given this specific sample.
- **Boundary Violations:** Asymptotic Wald intervals ($\widehat{N} \pm 1.96 \cdot \mathrm{SE}$) extend below m, placing mass on physically impossible values ($N < m$).
- **Inflexible to Decision Costs:** Cannot adapt to asymmetric operational losses where underestimation carries disastrous strategic costs.

Transition to Bayes: *Point estimators provide only an average; Bayesian inference constructs the full posterior distribution $P(N \mid D)$.*

SECTION 01

Theoretical Properties & Bayesian Foundations

Inferring the posterior distribution, reference priors, loss geometry, and boundary admissibility

Bayesian Inference: What Are We Trying to Infer?

Establishing the inferential objective: moving from raw identifiers to an epistemic probability distribution.

1. What Do We Know?

- The observed sample of serial numbers $D = \{s_1, \ldots, s_k\}$.
- The sample maximum $m = \max(s_1, \ldots, s_k)$.
- Physical fact: the population must satisfy $N \geq m$.

2. What Is Unknown?

- The true population ceiling $N \in \{m, m+1, m+2, \ldots\}$.
- We do not know how far above m the true ceiling lies.

3. What Does Bayesian Inference Give Us?

A posterior distribution describing the probability of every possible population size, conditional on the observed data:

$$\boxed{P(N = n \mid D)}$$

4. What Do We Do With This Distribution?

- Quantify uncertainty via exact credible intervals.
- Derive optimal point estimates tailored to specific loss functions.
- Make principled decisions accounting for risk asymmetry.

Inferential Pipeline:

Observed Data $D = \{s_1, \ldots, s_k\}$ $\longrightarrow$ **Posterior Distribution** $P(N = n \mid D)$ $\longrightarrow$ Estimation & Decision Rules $\widehat{N}_L$

Core Message: *The posterior distribution itself is the primary inferential output; point estimates are derived afterwards.*

Ingredients of Bayesian Inference: Likelihood, Prior, Posterior

How observed evidence combines with prior assumptions to quantify uncertainty about N.

1. Likelihood: Data Evidence

What does the sample tell us about N?

$$P(D \mid N = n) = \frac{1}{\binom{n}{k}} \mathbb{I}(n \geq m)$$

- Measures the probability of observing D under each candidate n.
- Equals **zero for** $n < m$: automatically assigns zero posterior probability to physically impossible population sizes.

2. Prior: Initial Assumptions

What do we assume about N before observing data?

- **Flat Prior ($\pi \propto 1$):** Assumes equal prior weight across all candidate integers.
- **Reciprocal Prior ($\pi \propto 1/n$):** Discrete analog of scale invariance; assigns equal mass to multiplicative decades/octaves ($\sum_{A}^{cA} \frac{1}{n} \approx \log c$).

3. Posterior: Updated Belief

What do we believe about N after observing data?

$$P(N = n \mid D) = \frac{\binom{n}{k}^{-1} \pi(n)}{\sum_{j=m}^{\infty} \binom{j}{k}^{-1} \pi(j)}$$

- Combines serial evidence with prior structure.
- Quantifies epistemic uncertainty across all physically admissible $n \geq m$.

Posterior Interpretation: *The posterior combines sample data with prior assumptions to assign exact probabilities to all candidate population sizes.*

Discrete Posteriors: Propriety & Moment Existence Thresholds

Different prior assumptions produce different posterior distributions, even when the observed data remain unchanged.

Exact Discrete Posterior PMFs ($n \geq m$)

- **Flat Prior ($\pi \propto 1$):**

$$P(N = n \mid m, k) = \frac{k-1}{k} \frac{\binom{m-1}{k-1}}{\binom{n}{k}}$$

Normalizing constant telescopes exactly via negative hypergeometric summation.

- **Reciprocal Prior ($\pi \propto 1/n$):**

$$P(N = n \mid m, k) = \frac{1}{Z(m,k)} \frac{1}{n\binom{n}{k}}$$

$Z(m,k) = \sum_{j=m}^{\infty} \frac{1}{j\binom{j}{k}}$ converges for all $k \geq 1$ (no simple closed form; evaluated numerically).

Moment Existence Thresholds & Pareto Limit

Prior Distribution	Proper Posterior	Finite Mean $\mathbb{E}[N]$	Finite Var $\mathrm{Var}(N)$
Flat Prior ($\pi \propto 1$)	$k \geq 2$	$k \geq 3$ ($\mathbb{E} = 2m - 2$)	$k \geq 4$
Reciprocal Prior ($\pi \propto 1/n$)	$k \geq 1$	$k \geq 2$ ($\mathbb{E} \approx 2m$)	$k \geq 3$

Continuous Pareto Limit ($N \gg k$):

$$p(N \mid m, k) \approx \frac{(k-1)m^{k-1}}{N^k}, \quad N \geq m$$

- Cumulative CDF: $F(N) = 1 - (m/N)^{k-1}$.
- Median: $\widehat{N}_{\mathrm{med}} \approx m \cdot 2^{1/(k-1)}$.

Theoretical Takeaway: *Reciprocal prior guarantees posterior propriety for all $k \geq 1$, whereas flat prior requires $k \geq 2$ for propriety and $k \geq 3$ for finite mean.*

Estimating N from a Loss Function: Bayesian Decision Theory

A posterior distribution describes uncertainty, but a decision requires us to select a single estimate.

The Decision-Theoretic Framework

- The optimal estimate depends on the cost assigned to different estimation errors.
- Let $a \in \mathbb{N}_{\geq m}$ denote an estimation action.
- Loss function $L(N, a)$ quantifies the consequence of predicting a when true population is N.
- **Posterior Expected Loss (Risk):**

$$R(a) = \mathbb{E}[L(N, a) \mid D] = \sum_{n=m}^{\infty} L(n, a) P(N = n \mid D)$$

- **Optimal Bayes Estimator:**

$$\widehat{N}_L = \arg\min_a R(a)$$

Example: Squared-Error Loss (L_2)

When estimation errors are penalized quadratically:

$$L(N, a) = (N - a)^2$$

$$R(a) = \mathbb{E}[(N - a)^2 \mid D] = \mathrm{Var}(N \mid D) + (\mathbb{E}[N \mid D] - a)^2$$

Setting the derivative to zero yields:

$$\boxed{\widehat{N}_{L_2} = \mathbb{E}[N \mid D] \quad \text{(Posterior Mean)}}$$

- The posterior mean is not "the" Bayesian estimate; it is optimal specifically for squared-error loss.
- If errors carry asymmetric costs, a different estimator is required.

Core Rule: *Decisions follow inference: point estimators cannot be chosen without defining a loss function.*

Loss Functions & Bayesian Estimators: Matching Risk to Mission

Point estimators shift systematically with the decision-maker's loss geometry.

1. 0–1 Loss

$L(N, a) = \mathbb{I}(a \neq N)$

$\widehat{N} = \text{Mode} = m$

- Selects most probable single value.
- Severe negative bias (-16.7% at $k = 5$).

2. Absolute (L_1)

$L(N, a) = |N - a|$

$\widehat{N} = \text{Median}$

- Linear penalty on deviations.
- Robust against heavy Pareto tails (-0.9% bias).

3. Squared (L_2)

$L(N, a) = (N - a)^2$

$\widehat{N} = \mathbb{E}[N \mid D]$

- Quadratic penalty on large errors.
- Pulled rightward by skewed Pareto tail.

4. Check Loss L_q

$L_q = (N - a)(q - \mathbb{I}(N < a))$

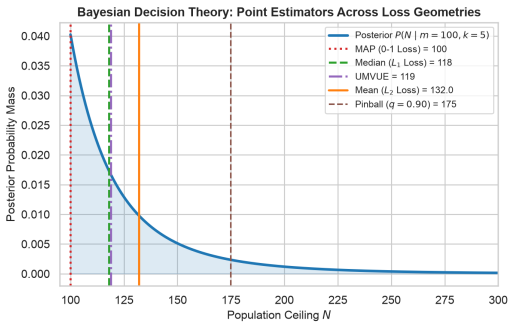
$\widehat{N} = F^{-1}(q \mid D)$

- $q = 0.5$: symmetric median.
- $q = 0.9$: undercounting penalized $9\times$ worse $\Longrightarrow$ defensive safety buffer.

Military Application: *In defense intelligence, underestimating enemy armor risks catastrophic defeat; asymmetric check loss ($q = 0.90$) provides a principled operational buffer.*

Numerical Illustration: Sensitivity of Estimates to Loss

Illustration using the flat-prior posterior, with sample maximum $m = 100$ and sample size $k = 5$.



Point Estimates Across Objectives ($m = 100, k = 5$)

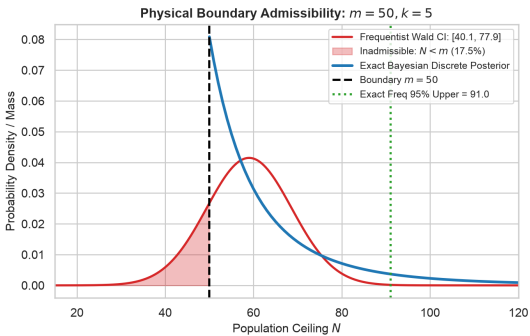
Objective / Loss	Estimator	Estimate	Strategic Meaning
0–1 Loss	MAP / Mode	**100**	Extreme lower bound; severe undercount
L_1 Absolute Loss	Posterior Median	**118**	Minimizes expected absolute error
L_2 Squared Loss	Posterior Mean	**132**	Quadratic balance; tail-inflated
Check Loss ($q = 0.90$)	90th Percentile	**175**	+75% defensive safety buffer

- Under the same observed data ($m = 100, k = 5$), optimal estimates span from 100 to 175.
- Posterior expected loss is minimized at a different point for each loss function.

Key Finding: *The same observed data can produce vastly different optimal estimates depending on how we value overestimation and underestimation.*

Physical Boundary Admissibility: Wald vs. Bayesian Intervals

The physical constraint $N \geq m$ must be respected by any valid uncertainty interval.



Frequentist Wald Boundary Collapse

Wald interval $\widehat{N} \pm 1.96 \cdot \mathrm{SE}$ yields $[40.1, 77.9]$ for $m = 50, k = 5$. It places **17.5% impossible mass** on $N < m = 50$. Asymptotic Gaussianity fails completely near the support boundary.

Approximate Frequentist Pivotal CI

Large-N pivotal interval $[m, m \cdot \alpha^{-1/k}] = [50.0, 91.0]$ respects the physical boundary by construction.

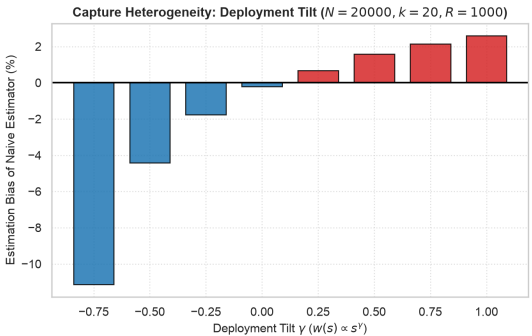
Bayesian Credible Coherence

Reciprocal-prior 95% credible interval strictly enforces physical admissibility ($P(N < m \mid D) = 0$) and approximately matches the pivotal bound $[50.0, 91.0]$.

Key Finding: *The reciprocal-prior posterior respects the physical lower bound and produces credible intervals that approximately agree with the large-N frequentist pivotal interval.*

Sensitivity to Non-Uniform Sampling: When the Maximum Misleads

Evaluating estimator robustness when capture probability varies with serial age: $w(s) \propto s^{\gamma}$.



Weighted Capture Mechanics

- $\gamma > 0$ (newer tanks prioritized): pushes $m \to N$, inducing $+1.6\%$ bias at $\gamma = +0.5$.
- $\gamma < 0$ (survivorship / depot attrition): induces -4.4% bias at $\gamma = -0.5$ and -11.1% bias at $\gamma = -0.75$.

Sufficiency Structure Breakdown

- **Known γ:** m remains sufficient for N via tilted factor $g_N(m; \gamma)$.
- **Unknown γ:** m alone ceases to be sufficient; joint minimal sufficient statistic is $(m, \sum_{i=1}^{k} \log s_i)$.

Methodological Warning: *Applying standard estimators when sampling is tilted introduces systematic bias; joint inference over (N, γ) is required.*

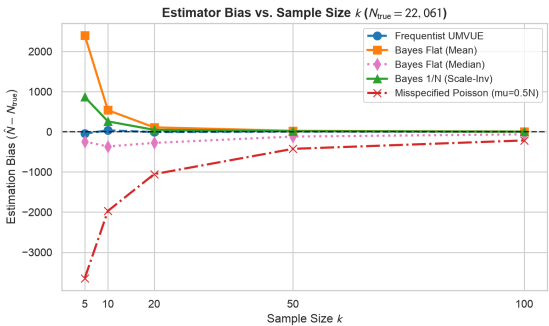
SECTION 02

Controlled Monte Carlo Simulation Study

Evaluating bias, RMSE contraction, interval calibration, and prior sensitivity across $R = 4,000$ trials ($N = 20,000$)

Estimator Bias: Finite-Sample Shrinkage vs. Unbiasedness

Evaluating mean bias convergence across $R = 4,000$ Monte Carlo replications on $N_{\text{true}} = 20,000$.



Frequentist UMVUE Unbiasedness

$\widehat{N} = m(1 + 1/k) - 1$ exhibits near-zero bias across all k (-48.0 ± 54.6 at $k = 5$, $+5.2 \pm 6.0$ at $k = 50$). Unbiased over hypothetical repeated samples.

Bayes Reciprocal Prior

Posterior Mean has modest bias ($+767.0 \pm 56.8$ at $k = 5$), rapidly decaying to $+13.2$ at $k = 50$. Posterior Median exhibits negative shrinkage (-901.5 at $k = 5$).

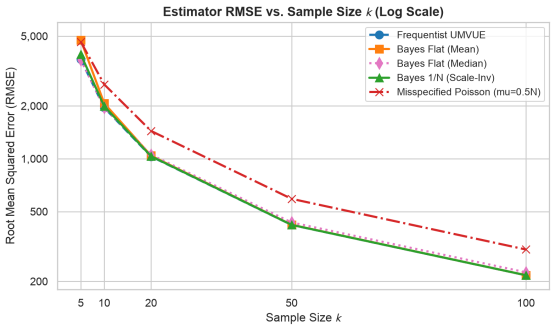
Flat Prior Tail Inflation

Skewed Pareto tail inflates flat posterior mean by $+2,168.6 \pm 60.7$ at $k = 5$.

Core Distinction: *Unbiasedness is a repeated-sampling frequentist property; Bayesian posterior summaries are chosen using the posterior distribution and a decision objective.*

Faster-Than-Standard-CLT Error Contraction: RMSE & Variance

Boundary order-statistic structure accelerates Root Mean Squared Error decay to $\mathcal{O}(k^{-1})$.



Theoretical Error Rate

Because N bounds the support, the UMVUE variance satisfies:

$$\mathrm{Var}(\widehat{N}) = \frac{(N-k)(N+1)}{k(k+2)} = \mathcal{O}(k^{-2})$$

giving accelerated rate RMSE $= \mathcal{O}(k^{-1})$, far outperforming standard CLT rate $\mathcal{O}(k^{-1/2})$.

Empirical Concordance

A $10\times$ increase in k ($5 \to 50$) yields a $\mathbf{9.0}\times$ error reduction ($3,455.2 \to 382.5$), closely matching theoretical $8.6\times$ factor and asymptotic $10\times$ scaling.

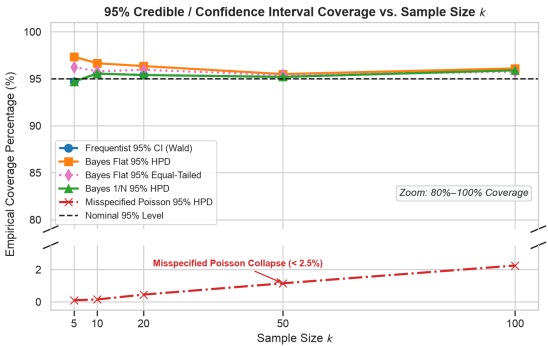
Estimator Convergence

For $k \geq 50$, all point estimators achieve identical precision (≈ 383).

Contraction Rule: *The order-statistic boundary structure drives an accelerated $\mathcal{O}(k^{-1})$ RMSE decay, reducing error by nearly an order of magnitude as sample size increases tenfold.*

Prior Sensitivity: Calibration and the Informative Prior Trap

Evaluating interval reliability and the catastrophic consequences of dogmatic prior misspecification.



Reciprocal Prior Calibration

Empirical coverage closely tracks nominal 95%: **94.4%** $\pm$ **0.36%** at $k = 5$ and **95.6%** $\pm$ **0.32%** at $k = 50$.

Informative Boundary Trapping

A dogmatically narrow misspecified Poisson($\mu = 10k$) prior traps posterior against the boundary $[m, m + 5]$, collapsing coverage to **0.3%**.

Hierarchical Hyperprior Resolution

Gamma hyperprior over Poisson rate ($\mathrm{CV} \geq 1$) provides adaptive dispersion, restoring coverage to **94.6%–95.0%** even under 50% prior mean error.

Calibration Finding: *The reciprocal prior produces empirical coverage close to 95% in our simulation. A misspecified Poisson prior causes severe collapse, while a hierarchical prior restores calibration.*

Monte Carlo Benchmark: Empirical Performance Leaderboard

Multi-dimensional evaluation across $R = 4,000$ replications on $N_{\text{true}} = 20,000$ (Seed 42).

Framework	Estimator	k	Mean Bias $\pm$ SE	RMSE (Emp / Th)	95% Cov $\pm$ SE	Mean Width
UMVUE (Frequentist)	$\widehat{N} = m(1 + 1/k) - 1$	5	-48.0 ± 54.6	3,455.2 / 3,380.3	90.0% $\pm$ 0.47% (Wald)	13,219
Approximate Pivotal CI	Large-N Pivotal Bound	5	—	—	**94.4%** $\pm$ 0.36%	13,644
Bayes Flat Prior	Posterior Mean	5	$+2,168.6 \pm 60.7$	4,409.0 / 4,363.9	**97.4%** $\pm$ 0.25%	18,535
Bayes Flat Prior	Posterior Median	5	-226.5 ± 54.1	3,431.3 / —	**97.4%** $\pm$ 0.25%	18,535
Bayes Reciprocal $(1/n)$	Posterior Mean	5	$+767.0 \pm 56.8$	3,671.1 / 3,618.6	**94.4%** $\pm$ 0.36%	13,622
Bayes Reciprocal $(1/n)$	Posterior Median	5	-901.5 ± 52.3	**3,427.5** / —	**94.4%** $\pm$ 0.36%	13,622
UMVUE (Frequentist)	$\widehat{N} = m(1 + 1/k) - 1$	50	$+5.2 \pm 6.0$	382.5 / 391.8	95.1% $\pm$ 0.34% (Wald)	1,536
Approximate Pivotal CI	Large-N Pivotal Bound	50	—	—	**95.6%** $\pm$ 0.32%	1,211
Bayes Reciprocal $(1/n)$	Posterior Mean	50	$+13.2 \pm 6.0$	382.8 / 392.0	**95.6%** $\pm$ 0.32%	1,209
Bayes Flat Prior	Posterior Mean	50	$+21.6 \pm 6.1$	383.4 / 392.4	**95.9%** $\pm$ 0.31%	1,237

1. Point Accuracy

UMVUE is exactly mean-unbiased; Bayes reciprocal median achieves lowest small-sample RMSE $(3,427.5)$.

2. Interval Reliability

Reciprocal credible interval achieves 94.4% coverage; Wald interval suffers from boundary distortion (90.0%).

3. Interval Precision

Reciprocal intervals are substantially sharper $(13,622)$ than flat prior intervals $(18,535)$.

Synthesis: *The reciprocal prior provides a useful balance between estimation accuracy, empirical calibration, and physical admissibility in the simulated setting.*

SECTION 03

Real-World Stress-Testing

Does the textbook German Tank model actually work on modern data?

The Investigation: Moving from clean mathematical simulations to messy operational reality. We stress-test textbook assumptions against historical WWII records and 541,909 modern e-commerce transaction lines.

The Premise: Spying on an Online Store

In WWII, Allies used serial numbers to count tanks. Can we do the same to count e-commerce sales?

The Observed Evidence

We collected an annual customer transaction ledger from a real e-commerce retailer:

- **541,909** individual item line purchases.
- Grouped into **22,061 distinct sales invoices**.
- Numbered sequentially from **536,365** to **581,587**.

The observed sequence span:

$$581{,}587 - 536{,}365 + 1 = \mathbf{45{,}223} \text{ slots}$$

The Immediate Mystery

If the store issued numbers between 536,365 and 581,587 (a span of 45,223 slots):

- Why are there only **22,061 active sales**?
- **23,162 numbers (51.2%)** are missing!

The Central Question:

The textbook assumes no gaps and random draws. What happens to our Bayesian estimates when applied to this messy operational stream?

Pedagogical Objective: We are not claiming commercial retail behaves like military tank manufacturing. We use this dataset to stress-test textbook assumptions: *how numbers are created, missing values, and what we are actually counting.*

Surprise 1: Numbers Flow in Time (Not Picked from a Hat)

Drawing balls from a static lottery jar is fundamentally different from watching a live stream.

Textbook Model: "The Static Hat"

- All N tanks exist in a warehouse before the spy begins sampling.
- An observer draws k numbers at random without replacement.
- Every tank has an equal, time-invariant probability of being observed.
- Order of observation carries no parameter information once sorted.

Operational Reality: "The River Stream"

- Invoices are generated sequentially on the calendar clock as customers buy.
- **Seasonal Arrival Acceleration:**
  - Q1 Baseline: $\approx$ **370** invoices/week.
  - Q4 Holiday Surge: $>$ **790** invoices/week.
  - Peak Week: **894** invoices in mid-November.
- High numbers (e.g., 580,000+) **did not exist** earlier in the year!

The Core Lesson: If you take a sample of invoices in March, you can never observe a high serial number. Sampling a live chronological process is **not** uniform random sampling from a static population.

Surprise 2: Half the Numbers Are Missing! (The 51.2% Void Mystery)

In the textbook, every number $1, 2, \ldots, N$ is a real tank. In e-commerce, 23,162 slots are empty.

1. Verified Cancellations (3,836 numbers)

- Account for **16.6% of all missing numbers**.
- **Confirmed in data:** Explicitly recorded with a `'C-'` prefix in the database (e.g., `C536379`).
- Represents customer returns, cancelled orders, and refunded transactions.
- These numbers were generated by the system, but the sale was reversed.

2. Unrecorded Gaps (19,326 numbers)

- Account for **83.4% of all missing numbers**.
- **Completely absent** from the transaction ledger.
- **Plausible Software Hypotheses:**
  - Abandoned online shopping carts.
  - Payment gateway timeouts before confirmation.
  - Clustered database servers reserving ID blocks that never completed.

The Core Lesson: In mechanical factories, stamping a serial number creates a physical tank. In modern enterprise software, **"number generated" does not mean "product sold"**.

Surprise 3: Evenly Spread Does NOT Mean Randomly Sampled

Why a smooth Kolmogorov–Smirnov test can trap an unsuspecting statistician.

The Tempting Test ($D \approx 0.012$)

- An analyst normalizes the observed invoice numbers into $[0, 1]$:

$$u_i = \frac{s_i - s_{\min}}{s_{\max} - s_{\min}}$$

- The empirical CDF is plotted against the uniform line $F(u) = u$.
- The maximum gap is tiny: $D = \sup |F_n(u) - u| \approx \mathbf{0.012}$.
- **Naive Temptation:** "The numbers look evenly spread across $[0, 1]$, so uniform sampling holds!"

Why That Logic Collapses

- **1. Statistical Trap:** Standard KS p-values are **invalid** because we estimated $s_{\min}$ and $s_{\max}$ from the data. The endpoints are forced to 0 and 1 by construction!
- **2. Conceptual Trap (Street Address Analogy):**
  - House numbers along a road $(1, 2, \ldots, 100)$ are spread perfectly evenly.
  - That does **not** mean packages delivered to houses were drawn from a random hat!

The Core Lesson: Goodness-of-fit measures spatial spread over the year, **not exchangeability or random sampling**. A steady river looks evenly spread along its banks, but it is still a flowing river.

The Big Trap: What Exactly Are We Trying to Count?

In the textbook, N is just "the number of tanks". In the real world, there are two different targets.

Target A: Sequence Capacity (N_{span})

- **Definition:** The total numeric address range opened by the database:

$$N_{\text{span}} = s_{\max} - s_{\min} + 1 = \mathbf{45{,}223}$$

- **What it tells you:** How high the company's internal software counter ticked.
- Includes empty slots, cancelled carts, and abandoned sessions.

Target B: Realized Sales (N_{count})

- **Definition:** The actual number of completed, paying customer orders:

$$N_{\text{count}} = \mathbf{22{,}061} \text{ sales}$$

- **What it tells you:** The commercial revenue volume of your competitor.
- Excludes all voided, cancelled, and empty sequence slots.

The $+105\%$ Catastrophe: If you blindly plug the observed invoices into the textbook formula $\widehat{N} = m(1 + 1/k) - 1$, it estimates the capacity ($\approx$ 45,223). If your executive thinks this represents customer sales, **your estimate is overstated by $+105\%$!**

Analogy: *It is like estimating hotel rooms (capacity) when you were asked to count the number of guests staying tonight.*

The Occupancy Model: The "Two Knobs" Dilemma

Connecting address capacity to active sales through an intuitive coin-flip framework.

The Coin-Flip Model (Thinning)

Imagine each of the N_{span} address slots becomes an active sale with probability ρ:

$$N_{\text{count}} \sim \text{Binomial}(N_{\text{span}}, \rho)$$

Across our complete annual dataset, the retrospective fill rate is:

$$\widehat{\rho} = \frac{22{,}061}{45{,}223} \approx \mathbf{0.4878} \quad (\mathbf{48.8\%})$$

The Relationship:

$\mathbb{E}[N_{\text{count}}] = \rho \times N_{\text{span}}$. If you know the capacity and the rate ρ, you can recover true sales!

The "Two Knobs" Dilemma

What if you only have a small sample of $k = 30$ invoices from a competitor?

- You observe gaps between numbers, but **two different worlds produce the exact same data**:
  1. Huge capacity ($N_{\text{span}} = 100{,}000$) with a low completion rate ($\rho = 22\%$).
  2. Moderate capacity ($N_{\text{span}} = 45{,}000$) with a moderate completion rate ($\rho = 49\%$).
- A small sample cannot turn both knobs at the same time!

The Takeaway: The occupancy model gives a sound theoretical bridge, but you **cannot estimate both capacity and fill rate from a small field sample alone** without an external audit of the company's software.

Section 3 Verdict: When Does the German Tank Model Work?

Answering the central question: bringing the empirical findings together.

The Scorecard: Comparing Two Worlds

WWII Tanks (Success)	Enterprise Streams (Failure)
Fixed monthly batches	Ongoing streaming transactions
Continuous sequential stamping	51.2% missing integer voids
Independent field captures	Changing seasonal arrival intensity
Single target (physical tanks)	Ambiguity: capacity vs. sales count
Statistical error: -4.4%	**Naive formula error:** $+105\%$

Three Golden Rules for Practitioners

- **1. Check How Numbers Are Made:** Determine whether serials come from a static batch or an evolving time-dependent stream.
- **2. Audit the Gaps:** Never assume missing identifiers represent active products without verifying cancellations or database block allocations.
- **3. Define What You Are Counting:** Explicitly choose between sequence capacity (N_{span}) and realized count (N_{count}) before running any estimator.

Final Verdict: The German Tank Problem works brilliantly when physical manufacturing matches textbook assumptions. In modern software and transactional systems, analysts must model the data-generating mechanism first.

Bottom Line: *Statistical elegance cannot rescue a model whose data-generating assumptions are violated.*

SUMMARY & CONCLUSIONS

Key Takeaways for Bayesian Discrete Inference

1. Objective Foundations

The target is the discrete ceiling N, with evidence captured by sample maximum m. Scale-invariant reciprocal prior $\pi(N) \propto 1/n$ yields proper posteriors ($k \geq 1$), respects physical boundaries ($N \geq m$), and matches the large-N pivotal bound.

2. Decision Theory & Rates

Point estimation follows loss geometry: median minimizes absolute error; check loss ($q = 0.9$) builds defensive buffers. Extreme-value order statistics produce accelerated $\mathcal{O}(k^{-1})$ RMSE contraction ($9\times$ reduction for $10\times$ sample).

3. Real-World Reality

Operational data violate textbook assumptions via chronological streaming and integer voids (51.2%). Bernoulli thinning disambiguates address capacity from realized transaction volume.

Central Research Answer: Bayesian inference provides a principled framework for quantifying uncertainty about an unknown discrete population ceiling, selecting estimates under mission-specific decision objectives, and evaluating inferential reliability. Its practical validity in the physical world, however, depends entirely on whether operational data-generation mechanisms satisfy model assumptions.

Thank You